

Leonardo de Andrade Santos

Electrical Engineering | Digital Predistortion, RF Power Amplifiers, FPGA, DSP and Digital IC Design

Master's researcher in Electrical Engineering investigating hardware-efficient Digital Predistortion (DPD) architectures for RF power amplifier linearization. Research focuses on behavioral modeling, fixed-point DSP, FPGA-oriented architectures, VHDL validation, and hardware-efficient algorithms for wireless transmitters.

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EDUCATION

M.Sc. in Electrical Engineering

Federal University of Paraná (UFPR), Brazil

03/2025 - Present

Research

- Research area: digital circuits and signal processing for wireless communications.
- Dissertation topic: hardware-efficient Digital Predistortion (DPD) architectures for RF power amplifier linearization.
- Investigated delay-dependent Memory Polynomial models for RF power amplifier behavioral modeling, targeting reduced computational complexity and hardware cost.
- Validated VHDL architectures with FPGA-oriented verification and hardware-cost analysis.
- Expected dissertation defense: August 2026.
- Supervisor: Prof. Eduardo Gonçalves de Lima.

B.Sc. in Electrical Engineering

Federal University of Paraná (UFPR), Brazil

03/2019 - 12/2024

Emphasis

- Academic emphasis on electronics, digital systems, embedded systems, signal processing, and integrated circuits.
- Teaching assistant in Digital Electronics and Microelectronics.
- Member of PET UFPR, contributing to university outreach, complementary technical education, and student-led academic projects.
- Member of the UFPR Baja SAE electrical subsystem, contributing to team-based engineering design and competitive academic activities.

RESEARCH INTERESTS

Digital Predistortion (DPD)

RF Power Amplifiers

Behavioral Modeling

Digital Signal Processing

FPGA-based Signal Processing

Digital IC Design

Wireless Communications

Hardware-Efficient Algorithms

PUBLICATIONS

Power Amplifier Modeling Using Memory Polynomials with Delay-Dependent Polynomial Orders (2026)

- Accepted for oral presentation at the 16th Workshop on Circuits and Systems Design (WCAS 2026).
- Proposed a delay-dependent Memory Polynomial variant for hardware-efficient DPD and RF power amplifier behavioral modeling.

- Evaluated trade-offs between modeling accuracy, polynomial order, memory depth, and hardware-oriented complexity.

M.Sc. Dissertation — Digital Predistortion for RF Power Amplifiers (2025–2026)

- Ongoing research on DPD behavioral modeling, fixed-point DSP, VHDL architectures, FPGA-oriented validation, and hardware-efficiency analysis.
- Investigates Memory Polynomial models, variable-order structures, LUT-based implementations, and accuracy-versus-cost trade-offs.

FULL CONFERENCE PAPERS

VHDL Description of a Memory Polynomial-Based Digital Predistorter (2024)

Lima, E. G.; Franca, S. B. L.; Santos, L. A. — SeMicro-PR, Curitiba, Brazil

- Published conference paper on the VHDL description of a Memory Polynomial-based digital predistorter.

Accuracy of Memory Polynomials with Fixed-Point Processing (2023)

Lima, E. G.; Franca, S. B. L.; Santos, L. A. — Paranaense Microelectronics Conference, Curitiba, Brazil

- Published conference paper evaluating fixed-point effects in Memory Polynomial-based modeling.

Design of a Dedicated Integrated Circuit for a Ring Oscillator-Based TRNG (2022)

Santos, L. A.; Franca, S. B. L.; Lima, E. G. — SeMicro-PR, Curitiba, Brazil

- Published conference paper on a dedicated digital IC design for a ring oscillator-based true random number generator.

ABSTRACTS IN CONFERENCE PROCEEDINGS

FPGA-Based Digital Predistorters for Wireless Transmitters (2023)

Santos, L. A.; Lima, E. G.; Franca, S. B. L. — 14th SIEPE, Curitiba, Brazil

Design of a Dedicated Integrated Circuit for a Ring Oscillator-Based TRNG (2022)

Santos, L. A.; Franca, S. B. L. — 13th SIEPE, Curitiba, Brazil

SELECTED ACADEMIC PRESENTATIONS

Paranaense Microelectronics Conference — Seminar Presentation (2024)

VHDL Description of a Memory Polynomial-Based Digital Predistorter

Paranaense Microelectronics Conference — Seminar Presentation (2023)

Accuracy of Memory Polynomials with Fixed-Point Processing

SIEPE — Seminar Presentation (2023)

FPGA-Based Digital Predistorters for Wireless Transmitters

Paranaense Microelectronics Conference — Seminar Presentation (2022)

Dedicated IC Design for a Ring Oscillator-Based TRNG

SIEPE — Seminar Presentation (2022)

Dedicated IC Design for a Ring Oscillator-Based TRNG

SIEPE — Seminar Presentation (2021)

BigPET: A Pedagogical and Scientific Experiment in Data Science and Machine Learning

SIEPE — Seminar Presentation (2021)

PET3D: Encouraging Maker Culture in PET

RESEARCH EXPERIENCE

Master's Researcher — Hardware-Efficient DPD and DSP

GICS-UFPR / Federal University of Paraná

03/2025 - Present

*Circuits and systems research for wireless communications
Research*

- Investigated RF power amplifier behavioral modeling using Memory Polynomial-based Digital Predistortion.
- Evaluated floating-point and fixed-point DSP models, including word-length optimization and integer-bound analysis.
- Proposed delay-dependent polynomial orders and evaluated 125 model configurations under Pareto trade-offs between accuracy and complexity.
- Validated VHDL architectures with pipelined datapaths and generate-based configurability.
- Validated numerical consistency between Python models and VHDL simulation results.
- Explored FPGA-oriented validation using Xilinx and HDL simulation tools.

Undergraduate Researcher — Digital Predistortion

GICS-UFPR / Federal University of Paraná

08/2022 - 12/2024

*DPD modeling, fixed-point processing, and hardware-oriented
implementation
Research*

- Investigated Memory Polynomial-based DPD for RF power amplifiers in undergraduate research.
- Evaluated fixed-point accuracy in Memory Polynomial models, resulting in a conference publication.
- Validated a VHDL digital predistorter for the undergraduate thesis, including digital IC implementation exploration.
- Presented modeling accuracy, fixed-point arithmetic, and hardware-cost trade-offs in academic venues.

Undergraduate Researcher — TRNG Integrated Circuit

GICS-UFPR / Federal University of Paraná

08/2021 - 08/2022

*Digital circuit design and integrated circuit implementation
Research*

- Investigated a ring oscillator-based True Random Number Generator (TRNG) for digital IC implementation.

- Evaluated and validated the architecture in VHDL and FPGA-oriented flows.
- Evaluated the circuit using a 130 nm BiCMOS 8HP technology flow.
- Applied Cadence NCLaunch, Genus, and Innovus for simulation, synthesis, and physical-design exploration.
- Validated output randomness using NIST statistical tests.

RESEARCH PROJECTS

Hardware-Efficient Digital Predistortion (2023–2026)

- Proposed and evaluated Memory Polynomial DPD models for RF power amplifier linearization, with emphasis on delay-dependent polynomial orders.
- Evaluated fixed-point DSP and VHDL architectures targeting FPGA-oriented validation and reduced hardware complexity.
- Connected behavioral modeling, numerical validation, and implementation-aware design for wireless transmitters.

TRNG ASIC (2021–2022)

- Investigated a ring oscillator-based TRNG as a digital IC design project in a 130 nm technology flow.
- Evaluated VHDL architecture, synthesis, physical-design exploration, and NIST statistical validation.

TEACHING & ACADEMIC ACTIVITIES

Teaching Assistant — Digital Electronics and Microelectronics

Federal University of Paraná (UFPR)

2021 - 2024

Undergraduate teaching support

Teaching

- Supported students in digital electronics, microelectronics concepts, laboratory activities, and technical exercises.
- Assisted practical activities involving digital logic, circuit analysis, and implementation-oriented topics.

Python Course Instructor

Federal University of Paraná (UFPR)

2021 - 2024

Complementary technical education for engineering students

Teaching

- Taught introductory Python programming for engineering applications.
- Covered programming fundamentals, problem solving, and practical computational examples.

Administrative Coordinator and Academic Project Member

SEATEL / PET UFPR

08/2021 - 12/2024

Student-led academic, outreach, and extension activities

Academic extension

- Coordinated courses, technical seminars, workshops, and academic extension activities.
- Contributed to IoPET, BigPET, and PET3D, emphasizing teamwork, university outreach, complementary education, and student leadership.

PROFESSIONAL EXPERIENCE

Software and Embedded Systems Engineer

IoTag Tecnologia

10/2022 - 02/2026
Precision agriculture and embedded IoT systems
Industry

- Contributed embedded firmware in C/C++ for ESP32/ESP8266-based IoT devices and agricultural machine interfaces.
- Maintained Python services and data tools using Flask/FastAPI, SQLAlchemy, PostgreSQL, Cassandra, AWS S3, and OpenCV.
- Worked with Linux-based development, embedded debugging, cloud integration, CI/CD, and deployment automation.

Engineering Intern

Lactec
06/2022 - 10/2022
Technology and research institute
Industry

- Supported the development of capacitive water-level and flow sensors for tidal energy systems.
- Contributed to electronics, microcontroller programming, sensor integration, and experimental validation.

ENGINEERING PROJECTS

UFPR Baja SAE – Electrical Subsystem (2019--2021)

- Contributed to a student engineering team through electrical design, wiring harnesses, telemetry support, firmware, PCB-related activities, integration, and vehicle testing.
- Participated in an academic competition environment emphasizing teamwork, leadership, documentation, and engineering excellence.
- Team results included 1st place in Electrical Design at the 2019 South Regional, 11th place nationally in 2020, and 2nd place in Electrical Design at the 2020 South Regional.

PET UFPR – Academic Extension Projects (2021--2024)

- Participated in IoPET, BigPET, and PET3D within a university extension program focused on complementary education and student-led technical development.
- Contributed to IoT, machine learning, 3D printing, course organization, outreach activities, and collaborative academic training.

TECHNICAL SKILLS

VHDLVerilogFPGA DesignDigital IC Design

ASIC FlowFixed-Point DSPDigital Predistortion

Memory PolynomialRF Power Amplifier Modeling

Behavioral ModelingPythonMATLABC/C++

Embedded LinuxESP32/ESP8266SQLAlchemyGit

CI/CDFastAPIFlaskPostgreSQLCassandra

AWS S3OpenCV

EDA TOOLS

Cadence Genus / Innovus / NCLaunchXilinx ISEVivado

ModelSim / QuestaGHDLGTKWaveYosysADS

Altium DesignerProteusLinuxNeovimVS Code

CERTIFICATIONS

Machine Learning with Python
60 hours

Databases and SQL
60 hours

Android Development
60 hours